MILITARY SPECIFICATION

CODER-BURST TRANSMISSION GROUP AN/GRA-71( )
(UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of equipment designated as Coder-Burst Transmission Group AN/GRA-71( ): (See 3.1 and 6.6).

Coder, Tape MX-4495( )/GRA-71
Coder, Tape MX-4496( )/GRA-71
Keyer KY-468( )/GRA-71
Magazine, Recording Tape MA-9( )/GRA-71
Adapter, Keyer MX-4498( )/GRA-71

1.2 Only those units specified in the invitation for bids or request for proposal shall be furnished by the contractor.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, flat; Steel
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock Corner
PPP-B-636	Boxes; Fiber
PPP-B-640	Boxes; Folding, Fiberboard, Corrugated, Triple Wall
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

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SPECIFICATIONS-(Cont'd)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-C-3885	Cable Assemblies and Cord Assemblies Electrical (For Use in Electronic, Communication, and Associated Electrical Equipment)
MIL-L-6085	Lubricating Oil: Aircraft Instrument, Low Volatility
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-46058	Coatings, Conformal (For Coating Printed Wiring Assemblies)
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Inspection Terms and Definitions
MIL-STD-129	Marking for Shipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Descriptions.-

3.1.1 Coder, Tape MX-4495( )/GRA-71.- This unit is a manual operated device that places an International Morse Code message into the Magazine, Recording MA-9( )/GRA-71. It contains three keys (DOT, DASH, and SPACE) which the operator presses to place the appropriate symbol on the tape. Pressing the dot and dash keys operates the tape drive advance, trips an impulse generator, and switches the impulse generator output to one of the windings of a dual track magnetic record head. Pressing the space key operates the tape drive advance. Pulses are placed on one track for dots and on the other track for dashes. No battery or external power is required.

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3.1.2 Coder, Tape MX-4496( )/GRA-71.- This unit produces Morse Code on Magnetic tape by single letter selection. This is accomplished by indexing a letter or a character wheel and then operating a printing ball. Tape advance is automatic for each character and word spacing is achieved by pressing the "word-space button." The character wheel contains the English alphabet, but no numbers. It is provided with a special dial which facilitates certain message processing functions. The message is on magnetic tape stored in Magazine, Recording Tape MA-9( )/GRA-71. After encoding is completed, the tape in the magazine is automatically rewound to its starting point by removal of the magazine. No batteries or external power is required.

3.1.3 Keyer KY-468( )/GRA-71.- This unit is a spring powered keyer designed for use with Magazine, Recording Tape MA-9( )/GRA-71. It uses a spring wound motor to drive the tape mechanism within the magazine. A crank which folds out of the way when not in use, is rotated to wind the motor. A connector on the bottom of Keyer KY-468( )/GRA-71 accepts a cable from the Adapter, Keyer MX-4498( )/GRA-71 or to an appropriate transmitter. Twelve volts DC, to operate the transistor circuitry within the Keyer, is furnished to this connector. A spring-return IDY button is provided to permit an operator to send a continuous stream of dots at a rate of 300 wpm. The IDY switch shall be locked out when the "ON-OFF" switch is in "ON" position. An inter-locked erase button is provided so that the Magazine, Recording MA-9( )/GRA-71 tapes can be erased by the keyer. The ON-OFF switch serves to start and stop the mechanical motor of the keyer. The switch is thrown to the ON position for message transmission and for erase.

3.1.4 Magazine, Recording Tape MA-9( )/GRA-71 .- This unit is a magnetic tape cartridge used to store at least 125 5-letter groups for transmission by Keyer KY-468( )/GRA-71. It contains 12-1/2 feet of high output instrument grade 1/4 inch magnetic tape. A rewind spring in the magazine automatically returns the tape to the beginning of the message wherever the magazine is removed from the coder or keyer.

3.1.5 Adapter, Keyer MX-4498( )/GRA-71.- This unit permits 300 wpm keying of the transmitter and provides watertight storage for Keyer KY-468( )/URT, Coder Tape MX-4495( )/GRA-71, Coder Tape MX-4496( )/GRA-71 and two Magazines, Recording Tape MA-9( )/GRA-71. This unit derives its input from the transmitter filament supply which may be 6.3 V RMS, 50 to 400 cps; or 6.3 Volts DC and converts it to 12 V DC at 50 Ma. to power Keyer KY-468( )/GRA-71, and includes provision for automatically turning off the transmitter oscillator between transmissions. The key adapter is housed in a moisture-proof case.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples for approval, if required by the invitation for bids or request for proposal, and contract. (See 4.3).

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3.3 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model, and the equipment shall incorporate all features of the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.4 Marking.-

3.4.1 General.- Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be Group 1 as described in that specification.

3.4.2 Serial numbers.- All items listed in 1.1 shall have serial numbers.

3.5 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after assembly. The final paint film shall be black color, semigloss, matching color chip 27038, in accordance with Federal Standard No. 595.

3.6 Lubricant.- Required lubrication of units shall be made using lubricating oil conforming to Specification MIL-L-6085.

3.7 Cable assemblies.- Cable assemblies shall conform to the requirements of Specification MIL-C-3885.

3.8 Printed wiring.- Printed wiring assemblies shall conform to Specification MIL-P-55110.

3.8.1 Conformal coating.- Conformal coatings used on printed wiring assemblies shall be Class A, Type ER of Specification MIL-C-46058.

3.9 Moisture and fungus treatment.- Units shall be treated to resist moisture and fungus in accordance with Specification MIL-T-152. Type II treatment shall apply.

3.10 Cleaning.-

3.10.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.10.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.11 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misalined condition.

3.12 Magazine, Recording Tape MA-9( )/GRA-71. -

3.12.1 The cartridge shall contain 12-1/2 feet of 3M Instrumentation Type 428 Magnetic Tape (formerly Type 128) of 1/4 inch width, 1.5 Mil polyester base, and "High Output" coating, or equal.

3.12.2 When tested as specified in 4.8.2, the following requirements shall be met:

(a) The MA-9( )/GRA-71 shall seat easily, lock firmly, shall not rock and shall release easily from the test fixture.

(b) The left tape spool (puller) shall have a minimum of 1-1/3 turns of tape when the cartridge is at rest.

(c) The right tape spool (supply) shall have a minimum of one turn, and the attaching tape shall not be visible.

(d) The MA-9( )/GRA-71 shall wind and rewind smoothly on the test fixture.

(e) The allowable wobble of either tape spool shall not exceed ± .002 inch maximum total.

(f) Neither binding nor sluggish action shall be detectable during rewind.

(g) A fully wound MA-9( )/GRA-71 shall rewind smoothly and quietly, in any position or attitude of the cartridge, when removed from the test fixture or keyer. Complete rewind shall occur in 3-1/2 seconds.

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3.12.3 When tested as specified in 4.8.3, the following requirements shall be met:

(a) The tape shall provide exact reproduction.

(b) The running time of the cartridge shall be at least 28 seconds, but not more than 32 seconds before the tape stops.

(c) Extension of the tape loop from the cartridge shall be possible for a distance of 6 feet ± 2 inches from the cartridge.

(d) Force at the periphery of the drive gear shall be between 35 to 65 grams inclusive.

(e) Dynamic tape tension shall not very outside of 3 to 12 grams over the entire length of tape travel while the tape is rewound into the cartridge, and 12 to 30 grams while the tape is pulled from the cartridge.

3.13 Coder, Tape MX-4495( )/GRA-71.-

3.13.1 Coder, Tape MX-4495( )/GRA-71, when tested as specified in 4.9.1, shall meet the following requirements:

(a) The depression of either "dot" or "dash" key shall operate generator ratchet one tooth. Generator output shall be four volts or greater.

(b) The recording head shall be aligned so that the direction of tape travel is perpendicular to the head gaps (azimuth).

(c) The force required to operate the "dot" button shall be between 20 to 35 oz.

(d) The force required to operate the "dash" button shall be between 20 to 35 oz.

(e) The force required to operate the "space" button shall be between 8 to 18 oz.

3.13.2 Magazine, Recording Tape MA-9( )/GRA-71, when encoded using Coder, Tape MX-4495( )/GRA-71 as specified in 4.9.2, shall be inspected for exactness of reproduction in comparison with a master tape.

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3.14 Coder, Tape MX-4496( )/GRA-71.-

3.14.1 Coder, Tape MX-4496( )/GRA-71, when tested as specified in 4.10.1, shall meet the following requirements:

(a) Operation of the handle, required for character generation, shall be smooth with no detectable binding throughout the entire movement. The force required for unextended handle "break-away" at points along arc of travel shall not exceed 20 oz. at room temperature. (Average of three values obtained on same unit).

(b) The force required to operate the word space button for operation of the space advance mechanism shall average within 36 ± 4 oz. at room temperature. (Average of three values obtained on same unit).

(c) Force to operate code dial shall be 4 ± 1 oz. in either direction, at room temperature.

3.14.2 Magazine, Recording Tape MA-9( )/GRA-71, when encoded using Coder, Tape MX-4495( )/GRA-71 as specified in 4.10.2, shall be inspected for exactness of reproduction in comparison to a master tape and the following characteristics:

(a) Dash-Dot width relationship: 3 to 1 (dot equals one baud).

(b) Pulse Shape (reasonably square) at the output of the keyer.

(c) Word Space (7 bauds), Character Space (3 bauds), Element (or within character space (1 baud)).

(d) Message capacity of cartridge (4,500 dots or 9000 bauds equivalent to 150 cps dot frequency.)

3.15 Keyer KY-468( )/GRA-71 (see 4.11).-

3.15.1 Keyer KY-468( )/GRA-71, when tested as specified in 4.11.1, shall meet the following requirements:

(a) Gears shall be properly aligned and the gear train shall be free running.

(b) When the slide button is pushed to "OFF" position, some resistance to the motion of the slide button shall be detected at least 1/32 inch before the slide button reaches its stop.

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(c) During wind-up in normal direction (clockwise), the keyer motor shall wind smoothly, and shall not attempt to unwind when winding is stopped and crank is released.

(d) The brake shall not slip at full wind unless an attempt is made to overwind the motor. The brake shall slip to prevent overwind but shall not slip enough to allow the spring to start unwinding.

(e) The governor shall meet the following requirements:

(1) With wind-up crank turned one or two turns, and with slide button in "ON" position, governor assembly shall slow down, stop, and tend to rotate in a reverse direction before coming to final stopped position.

(2) The governor shall rotate at 2520 ± 70 rpm and the speed shall remain constant through at least 75% of run-down time.

(3) The spring shall have capacity for two consecutive cartridge runs from full wind to complete run-down.

(4) Force at the periphery of output gear shall exceed 200 grams through 75% of run-down time.

3.15.2 Keyer KY-468( )/GRA-71, when tested as specified in 4.11.2, shall meet the following requirements:

(a) IDY and erase slide buttons shall operate easily and return to "OFF" positions automatically when released. Case shall not interfere with button operation. IDY and erase interlocks shall be operative.

(b) Gear train shall be free running with minimum backlash. Turn crank 1/2 turn and allow keyer to run down while observing output gear. Output gear shall stop and then reverse rotation for at least a fraction of a turn.

(c) Functional requirements:

(1) IDY frequency shall be 150 cps, plus or minus 5 cps.

(2) IDY on-off ratio shall be 50%, plus or minus 3%.

(3) Dot duration shall be 3.33 milliseconds, plus or minus 0.17 millisecond.

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(4) Dot frequency shall be 150 cps plus or minus 5 cps.

(5) Dash duration shall be 10.0 milliseconds, plus or minus 0.5 milliseconds.

(6) Space duration within characters shall be 3.33 milliseconds, plus or minus 0.17 milliseconds average over the length of the tape with 3 readings of the tape.

(7) Space duration between characters shall be 10.0 milliseconds, plus or minus 0.5 milliseconds average over the length of the tape with 3 readings of the tape.

(8) Peak-to-peak voltage of pulse train shall be essentially the same as keyer line supply voltage.

(9) Current drain of keyer electronic peakage (as read on ammeter) shall not exceed 40 ma.

(10) Erase function shall operate properly.

3.16 Keyer Adapter MX-4498( )/GRA-71. - When tested as specified in 4.12, the following requirements shall be met:

(a) Switch 1 and 2 to ON, and switch 3 to "A". - Measure input current to Keyer Adapter on A. Acceptable values 350 to 700 milliamperes AC.

(b) Screen grid regulator shall clamp at nominal 200 VDC at Test Point 6.

(c) Switch 3 at A. Nominal 100 VDC at Test Point 1 indicates application of proper voltage to PA keying circuit.

(d) Switch 3 at A. Nominal 40 VDC at Test Point 2 indicates application of proper voltage to oscillator keying circuit.

(e) Switch 3 at "B", and scope at Test Point 1. Pulses of 90 to 105 volts peak-to-peak with nominal 3.3 millisecond pulses should be observed. Also check at Test Point 2 for close to zero voltage on OSC cathode output.

(f) Switch 3 to "A". Observe rise time at Test Point 2. Acceptable values .3 to 1.0 seconds.

(g) Switch 3 at "A". Measure voltage to Test Point 3. Acceptable value, 10.8 to 14.0 VDC. (Minus-to-ground).

(h) Switch 3 to "A". Position and measure peak-to-peak ripple voltage at Test Point 3. Maximum, as observed on an oscilloscope, shall not exceed 0.16 volts.

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3.17 Service conditions.- The equipment shall meet the following service conditions requirements:

3.17.1 Immersion (Adapter, Keyer MX-4498( )/GRA-71 only).- Three feet of water for 2 hours (See 4.13.1).

3.17.2 Temperature.- All units of Coder-Burst Transmission Group AN/GRA-71( ) shall be capable of withstanding storage temperatures of -40°C and + 66°C for one (1) hour, and shall meet the operation inspection (4.17) at any temperature in the range of -20°C to +55°C.

3.17.3 Vibration (Keyer KY-468( )/GRA-71 only).- Three (3) G's at 30 cps for 10 minutes in each of three mutually perpendicular planes.

3.18 Interchangeability.- Like units, assemblies, subassemblies, modules, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.18). Individual items shall not be handpicked, for fit or performance; however, matched pairs or sets when permitted, may be interchangeable as such.

3.19 Equivalency.- Operational comparison between the procurement model and fabricated equipment shall be made to insure that the equipment on order performs at least equally as well as the procurement model.

3.20 Compatibility.- The interchange of fabricated units of Coder-Burst Transmission Group AN/GRA-71( ) with units of the procurement model shall not degrade the performance below that obtained using the procurement model only.

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3.21 Operational life tests (see 4.16). - Units of Coder-Burst Transmission Group AN/GRA-71( ) shall be subjected to the following operational life tests:

a. Coder, Tape MX-4495( )/GRA-71. -

	Preproduction Tests	Production Tests	
(1) Depress "DOT" key	5000	1500	times.
(2) Depress "DASH" key	5000	1500	times.
(3) Depress "SPACE" key	1500	500	times.

(4) The coder shall meet tne requirements of 3.13 following the life test.

b. Coder, Tape MX-4496( )/GRA-71. -

(1) The handle of Coder, Tape MX-4496( )/GRA-71 shall be depressed:

(a) (Preproduction Tests) 50 times for each character, in sequence, and then the "WORD-SPACE" button shall be depressed 300 times.

(b) (Production Tests) 25 times for each character in sequence, and then the "WORD-SPACE" button shall be depressed 150 times.

(2) The Coder shall meet the requirements of 3.14 following the life test.

c. Magazine, Recording Tape MA-9( )/GRA-71 and Keyer KY-468( )/GRA-71. - Magazine, Recording Tape MA-9( )/GRA-71 shall be wound on Keyer KY-468( )/GRA-71 and unwound 500 times for preproduction tests, and 300 times for production tests. Following the life tests, the magazine shall meet the requirements of 3.12.2 and 3.12.3, and the Keyer shall meet the requirements of 3.15.

3.22 Operational inspection. - The operational inspection specified in 4.17 shall be performed to determine that each unit is operable prior to packaging.

3.23 Bounce preconditioning. - The units shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.24 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.25 Workmanship. - Units, subassemblies and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirement.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery.) (See 4.3).

(b) Inspection covered by subsidiary documents. (See 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.20).

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and group C (see Tables II, III, and IV respectively).

Table I.- Preproduction inspection.

Inspection	Req para.	Insp para.
Equivalency	3.19	4.14
Compatibility Test	3.20	4.15
Temperature Test	3.17.2	4.13.2
Vibration Test	3.17.3	4.13.3

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Marking	3.4
Finish, protective	3.5
Lubricant	3.6
Cable assemblies	3.7
Printed wiring	3.8
Moisture and fungus treatment	3.9

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government-- at its discretion-- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection, except preproduction samples, shall be preconditioned after final assembly. (See 3.23).

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government except that operational inspection (4.17) shall be last. (See 6.5).

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Table II.- Group A inspection.

Inspection	Req para.	Insp para.	AQL Major	AQL Minor
Visual and Mechanical Inspection	3.25	4.19		
Coder, Tape MX-4495( )/GRA-71			1%	4 dphu⊛
Coder, Tape MX-4496( )/GRA-71			1%	4 dphu
Keyer, KY-468( )/GRA-71			1%	4 dphu
Magazine, Recording Tape MA-9( )/GRA-71			1%	4 dphu
Keyer, Adapter, MX-4498( )/GRA-71			1%	4 dphu
Magazine, Recording Tape MA-9( )/GRA-71				
Cartridge tape	3.12.1	4.8.1	2.5 dphu	⊛⊛
Seating, locking, rocking and releasing	3.12.2(a)	4.8.2(a)		
Left spool (turns)	3.12.2(b)	4.8.2(b)		
Right spool (turns) and attaching tape visibility	3.12.2(c)	4.8.2(c)		
Wind and rewind	3.12.2(d)	4.8.2(d)		
Wobble	3.12.2(e)	4.8.2(e)	4.0 dphu	
Rewind (binding and sluggish action)	3.12.2(f)	4.8.2(f)	for the	⊛⊛
Rewind of fully wound cartridge	3.12.2(g)	4.8.2(g)	group	
Exactness of reproduction	3.12.3(a)	4.8.3(c)	combined	
Running time of cartridge	3.12.3(b)	4.8.3(d)		
Extension of tape loop	3.12.3(c)	4.8.3(e)		
Force at periphery of drive gear	3.12.3(d)	4.8.3(f)		
Dynamic tape tension	3.12.3(e)	4.8.3(g)		
Inspection for physical and functional interchangeability	3.18	4.18	2.5 dphu	⊛⊛
Operational inspection	3.22	4.17	1%	⊛⊛
Coder, Tape MX-4495( )/GRA-71				
General operation and output	3.13.1(a)	4.9.1(a)		
Recording head alignment	3.13.1(b)	4.9.1(b)		
Force required for "dot" button	3.13.1(c)		4.0 dphu	
Force required for "dash" button	3.13.1(d)	4.9.1(c)	for the	
Force required for "space" button	3.13.1(e)		group	
Exactness of reproduction	3.13.2	4.9.2(j)	combined	
Inspection for physical and functional interchangeability	3.18	4.18	2.5 dphu	⊛⊛
Operational inspection	3.22	4.17	1%	⊛⊛

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Table II. - Group A inspection-(continued)

Inspection	Req para.	Insp para.	AQL Major	AQL Minor
Coder, Tape MX-4496( )/GRA-71				
Handle operation and force required	3.14.1(a)	4.10.1(a))	4.0 dphu	⊛⊛
Word space button tension	3.14.1(b)	4.10.1(b))	for the	
Force to operate code dial	3.14.1(c)	4.10.1(c))	group	
Exactness of reproduction and other characteristics	3.14.2	4.10.2(1))	combined	
Inspection for physical and functional interchangeability	3.18	4.18	2.5 dphu	⊛⊛
Operational inspection	3.22	4.17	1%	⊛⊛
Keyer KY-468( )/GRA-71				
Gear alignment and gear train operation	3.15.1(a)	4.11.1(a))		
Resistance to slide button motion	3.15.1(b)	4.11.1(b))		
Wind-up	3.15.1(c)	4.11.1(c))	4.0 dphu	⊛⊛
Brake slippage	3.15.1(d)	4.11.1(d))	for the	
Governor assembly:	3.15.1(e)	4.11.1(e))	group	
Operation	3.15.1(e)(1)	4.11.1(e)(1))	combined	
Rotation speed and speed constancy	3.15.1(e)(2)	4.11.1(e)(2))		
Spring capacity	3.15.1(e)(3)	4.11.1(e)(3))		
Force at periphery of output gear	3.15.1(e)(4)	4.11.1(e)(4)		
Assembled KY-468( )/GRA-71	3.15.2	4.11.2)		
Slide button operation	3.15.2(a)	4.11.2(a))		⊛⊛
Gear train operation	3.15.2(b)	4.11.2(b))	4.0 dphu for	
Functional requirements	3.15.2(c)	4.11.2(c))	group combined	
Operational inspection	3.22	4.17	1%	⊛⊛
Inspection for physical and functional interchangeability	3.18	4.18	2.5 dphu	⊛⊛
Keyer, Adapter MX-4498( )/GRA-71	3.16	4.12	4.0 dphu	⊛⊛
Inspection for physical and functional interchangeability	3.18	4.18	2.5 dphu	⊛⊛
Operational inspection	3.22	4.17	1%	⊛⊛

⊛ dphu = Defects per hundred units.

⊛⊛ All defects in these categories are considered major.

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5% defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III.- Group B inspection.

Inspection	Req para.	Insp para.	AQL (See 4.5.2)
Immersion	3.17.1	4.13.1	

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and normally be performed on sample units that have been subjected to and met group A and B inspection.

Table IV. - Group C inspection.

Inspection	Req para.	Insp para.
Operational life test [1]	3.21	4.16
[1] Units subjected to operation life tests shall not be furnished on the order.		

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4.5.3.1 Sampling for inspection of equipment.- One unit of each type on order shall be selected from the first quantities produced; thereafter one unit of each type from each 250, or fraction thereof produced.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of products which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4).

4.6 Bounce preconditioning.- The units of Coder-Burst Transmission Group AN/GRA-71( ) placed in their normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal, The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The units shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the units shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to Quality Conformance Inspection.

4.7 Test requirements.-

4.7.1 Test facilities.- Except for those items specifically indicated to be Government Furnished, in the Invitation for Bid, or Request for Proposal, and Contract, the contractor shall provide all test facilities and equipment required for preproduction and production tests. The test equipment and facilities shall be subject to the approval of the Contracting Officer.

4.7.2 The Government reserves the right to perform comparative tests (procurement model vs. fabricated equipment) using Government transmission and receiving facilities to insure compliance with specified requirements.

4.7.3 Standard message.- Where the use of a message is specified, the message to be coded shall include all the letters in the alphabet and the length of the message shall be equivalent to 100-five letter words.

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4.8 Magazine, Recording Tape MA-9( )/GRA-71.-

4.8.1 The cartridge shall be inspected for compliance with 3.12.1.

4.8.2 The following tests shall be performed using a "winding" fixture fabricated from a Coder, Tape MX-4495( )/GRA-71 or Coder, Tape MX-4496( )/GRA-71 head block including output gear and shaft with added drive gear shaft and crank:

(a) Check to insure that the MA-9( )/GRA-71 meets the seating, locking, rocking and releasing requirement of 3.12.2(a).

(b) Check for minimum number of turns on left tape spool (puller) visually by lifting tape off spool.

(c) Check for minimum number of turns and visibility of attaching tape by winding MA-9( )/GRA-71 to stop on fixture. Run cartridge to the end of travel on keyer and check reserve on right spool while holding thumb on left spool to keep from rewinding.

(d) Check MA-9( )/GRA-71 for smooth winding and rewinding.

(e) Check MA-9( )/GRA-71 for maximum wobble.

(f) Check to insure that neither binding nor sluggish action is detectable during rewind when a fully wound MA-9( )/GRA-71 is removed from test fixture or keyer and stopped every 1-1/2 to 3 turns of left tape spool (puller).

(g) Check to insure that a fully wound MA-9( )/GRA-71 meets the requirements of 3.12.2(a).

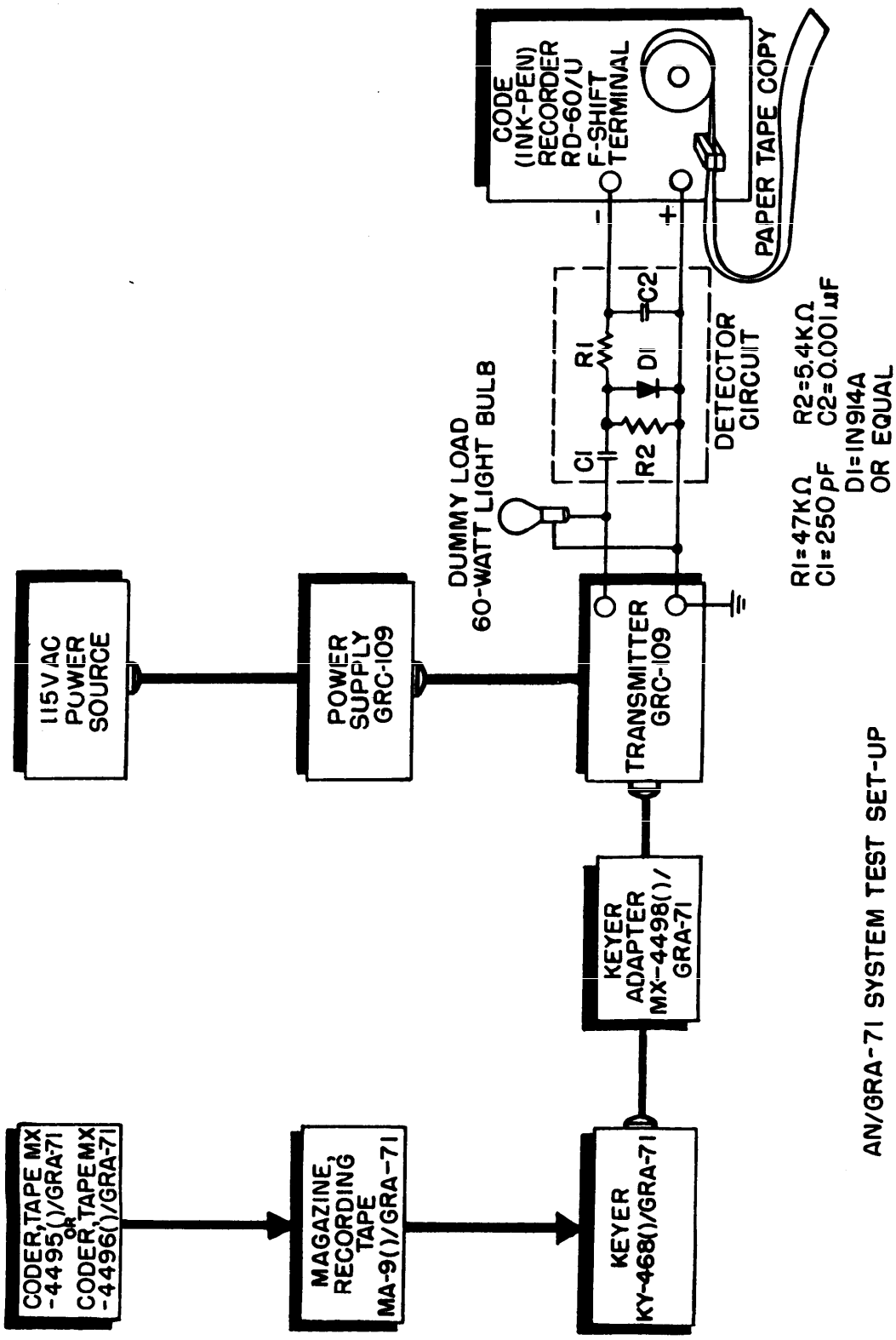
4.8.3 The following test shall be performed using a Coder, Tape MX-4495( )/GRA-71 or Coder, Tape MX-4496( )/GRA-71, Keyer KY-468( )/GRA-71, Radio Transmitter T-784( )/GRC-109, Coder Recorder RD-60/U, and detector circuit (see Figure 1):

(a) Polarize MA-9/GRA-71 tape by erasing the cartridge by means of Keyer KY-468/GRA-71.

(b) Encode a predetermined message upon the tape by use of either Coder MX-4495( )/GRA-71 or MX-4496( )/GRA-71.

(c) Play the tape through the Keyer and Transmitter assembly, and record the detected output on the RD-60/U. Read the tape and check for exactness of reproduction. The tape shall be free of errors.

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AN/GRA-71 SYSTEM TEST SET-UP

FIGURE 1

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(d) Note the running time of a cartridge from start to finish with fully wound Keyer.

(e) Extend tape loop from cartridge by means of a pencil or rod.

(f) Allow the tape cartridge to rewind. Check the force at the periphery of the drive gear using a gram scale.

(g) The dynamic tape tension shall be checked over the entire length of tape travel.

4.9 Coder, Tape MX-4495( )/GRA-71.- (See 3.13).

4.9.1 After assembly, but prior to attaching the case, inspect and test the MX-4495/GRA-71 for the following:

(a) Verify that depression of either dot or dash key operates generator rachet one tooth. Generator output shall be measured across dot and dash switches and observed on calibrated oscilloscope.

(b) The recording head alignment shall be checked. This shall be accomplished by an optical method employing a magnifier fixture with right angle markings within the field of view. A rough test for proper alignment can be made by playback of a recorded cartridge on the test setup employing the RD-60/U. A skewed head to tape condition is readily evidenced by noting a stretching or compressing of spaces between recorded dots and dashes.

(c) The force required to operate the "dot" button, "dash" button, and "space" button shall be measured by means of a tension gauge.

4.9.2 The following tests shall be performed using completely assembles coders:

(a) Attach an MA-9/GRA-71 that has been erased and has passed final inspection.

(b) Depress space key about ten times to move exposed tape past recording head.

(c) Code the tape as follows: 5 dashes, 2 spaces, and 5 dots coded at the rate of one key depression per second.

(d) Code the tape as follows: 5 spaces, 5 dashes, 2 spaces, and 5 dots coded at the rate of 2 key depressions per second.

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(e) Code the tape as follows: 5 spaces, 5 dashes, 2 spaces, and 5 dots coded at the rate of 3 key depressions per second.

(f) Code the tape as follows: space, dot, space, dash for 5 consecutive times at the rate of 1 key depression per second, 2 per second, and 3 per second.

(g) Repeat (a) through (f).

(h) Detach cartridge and allow it to rewind completely.

(i) Play cartridge into test fixture. Test fixture (Fig. 1) shall consist of KY-468/GRA-71 connected to appropriate transmitter, with transmitter working into a Code Recorder RD-60/U.

(j) Read the tape and check for exactness of reproduction.

4.10 Coder, Tape MX-4496( )/GRA-71 (See 3.14).-

4.10.1 After assembly, but prior to attaching the case, inspect and test the MX-4496( )/GRA-71 for the following:

(a) Check handle operation. The force required for unextended handle "break-away" shall be measured by means of a tension gauge (Chatillon Gauge-R, or equal).

(b) Check force required for operation of word space button by means of a tension gauge (Chatillon-R).

(c) Check force required for operation of the code dial in either direction at the periphery.

4.10.2 Attach case to assembled MX-4496/GRA-71 and proceed with final inspection as described below.

(a) Attach an MA-9/GRA-71 that has been erased and has passed final inspection.

(b) Depress word space button six times to move exposed tape past recording head.

(c) Rotate character dial to letter "A". (Red Scale).

(d) Grasp operating handle firmly between thumb and first three fingers. Raise handle up and back as far as it will go. Bring handle down with firm, even stroke until it stops. Repeat this operation three times at a rate of 30 strokes per minute.

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(e) Depress word space button two times.

(f) Rotate character dial to letter "B" and repeat step (d).

(g) Repeat step (e).

(h) Continue with above procedure until entire alphabet is recorded.

(i) Depress word space button ten times and repeat encoding of entire alphabet as described above, but at a rate of 60 strokes per minute.

(j) Detach cartridge from coder and allow it to rewind completely.

(k) Check tape encoding by playing cartridge into test fixture (Fig. 1). Test fixture shall consist of MX-4498/GRA-71 and KY-468/GRA-71 (procurement models from previous production) connected to appropriate transmitter, with transmitter working into Code Recorder RD-60/U.)

(l) Read the tape and check for exactness of reproduction and check the characteristics specified in 3.14.2(a) through 3.14.2(d).

4.11 Keyer KY-468( )/GRA-71 (See 3.15).-

4.11.1 After assembly, but prior to installation of the electronic package, inspect KY-468( )/GRA-71 for the following:

(a) Inspect gear alignment and check gear train operation.

(b) Verify that there is some resistance to pushing the slide button to the "OFF" position.

(c) Turn wind-up crank clockwise, stopping at various angles of rotation.

(d) Inspect for brake slippage.

(e) Governor assembly:

(1) Inspect for operation.

(2) Check Governor assembly rotation and speed constancy with a stroboscope.

(3) Check spring capacity.

(4) Check force at periphery of output gear.

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4.11.2 With electronics package installed and case attached, inspect KY-468( )/GRA-71 as follows:

(a) Check IDY and erase slide buttons for proper operation. Check for interference of slide buttons and case.

(b) Check gear train for proper operation.

(c) Check functional requirements by connecting keyer to keyer test set-up with Tektronics Type 181 Time Mark Generator. (See Fig. 2), and using a calibrated oscilloscope and a tape cartridge having series of dots and dashes recorded with and without spaces interposed. An electronic counter shall be used to determine compliance with 3.15.2(c)(1) and (4). An oscilloscope shall be used to determine compliance with 3.15.2(c)(2), (3), (5), (6), (7), and (9). Check erase function by erasing a previously recorded cartridge and playing erased cartridge on keyer under test. No output should appear on oscilloscope screen. Record cartridge again and play back into Fig. 1 test set-up. Recorded message shall appear on Recorder RD-60/U. Erase cartridge.

4.12 Keyer Adapter MX-4498( )/GRA-71.- The MX-4498/GRA-71 Keyer Adapter test set-up (Fig. 3), shall be used to perform the tests. Test fixtures shall consist of simulated KY-468/GRA-71 (3.3 millisecond shorting switch), power supply (6.3 VAC/DC, 500 VDC, and 100 VDC), and switching mechanism for connecting power source and keyer to MX-4498/GPA-71.

(a) Connect MX-4498/GRA-71 KEYER and TRANSMITTER Cables to test set-up.

(b) Using voltmeter and calibrated oscilloscope, test the functions in 3.16 for conformance with specified values.

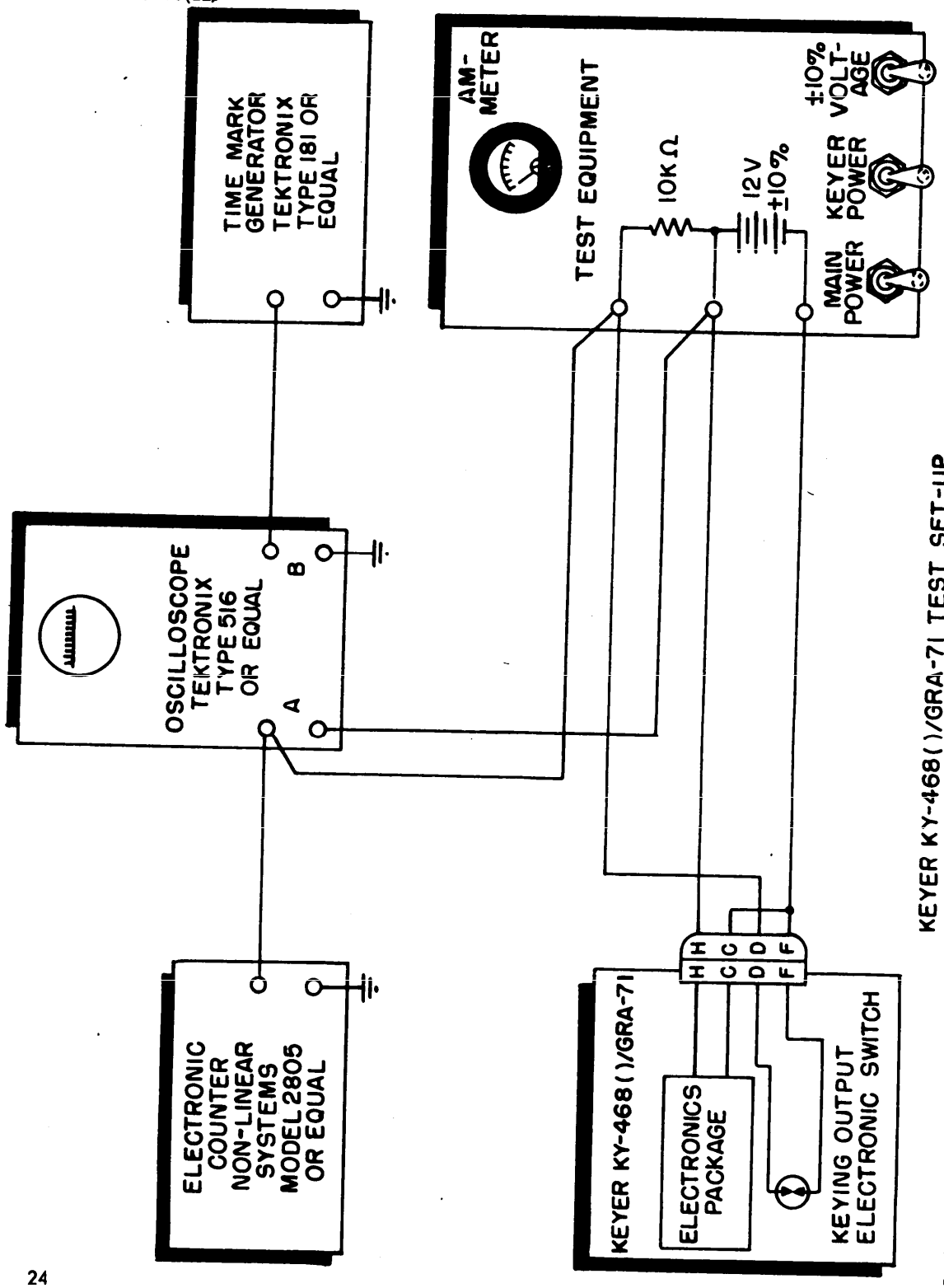
4.13 Service conditions.-

4.13.1 Immersion test.- Keyer Adapter MX-4498( )/GRA-71 shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of Keyer Adapter MX-4498( )/GRA-71 shall be 40°F, or more, above the temperature of the water. The tank in which the Keyer Adapter is immersed shall be of sufficient capacity to maintain the water within ± 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the Keyer Adapter MX-4498( )/GRA-71 shall be removed from the water and wiped dry on exterior surfaces. When the Keyer Adapters are opened, there shall be no evidence of leakage.

4.13.2 Temperature.- All units shall be tested for compliance with 3.17.2.

4.13.3 Vibration.- Keyer KY-468( )/GRA-71 shall be tested for compliance with 3.17.3.

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KEYER KY-468( )/GRA-71 TEST SET-UP

FIGURE 2

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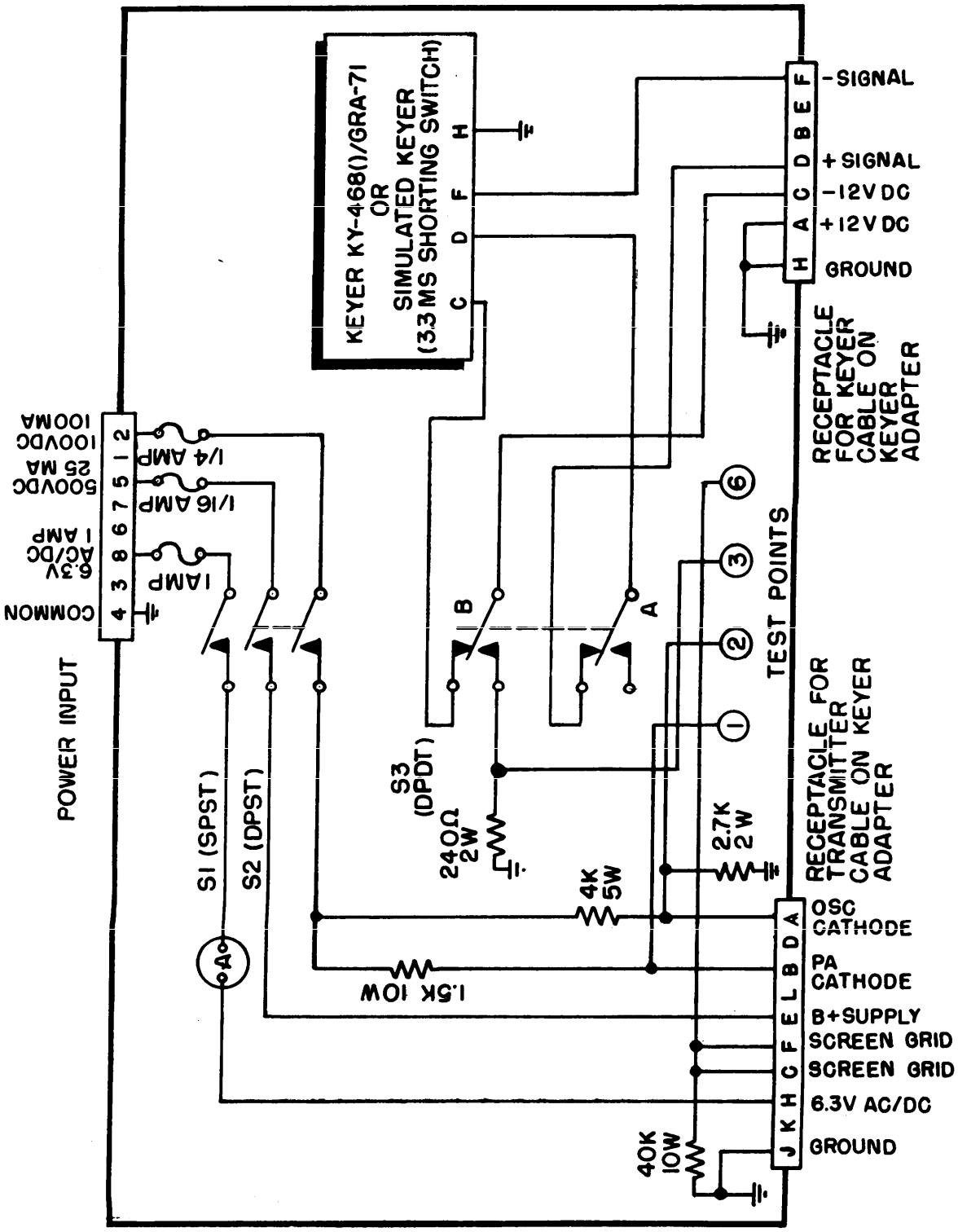


FIGURE 3. KEYER ADAPTER MX-4498()/GRA-71 TEST SET-UP

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4.14 Equivalency test.- Coder-Burst Transmission Group AN/GRA-71, produced on the order, shall be tested for comparative results against the procurement model. Specifically, identical message shall be coded on Magazine Recording Tapes MA-9( )/GRA-71 using Coder MX-4495( )/GRA-71 and Coder MX-4496( )/GRA-71 and the message shall be processed, using Keyer KY-468( )/GRA-71 and Keyer, Adapter MX-4498( )/GRA-71, over the test facility (Fig. 1) for compliance with 3.19.

4.15 Compatibility tests.- Units of Coder-Burst Transmission Group AN/GRA-71( ) produced on the order shall be interchanged with corresponding units of the procurement model in all combinations and coded messages shall be processed over the test facility (Fig. 1) for compliance with 3.20.

4.16 Operational life test.- Coder-Burst Transmission Group AN/GRA-71( ), or units thereof, as applicable shall be tested for compliance with 3.21.

4.17 Operational inspection.- Using the applicable test facility, units shall be checked and shall meet the following:

	Req.	Test
Magazine, Recording Tape MA-9( )/GRA-71	3.12.3(a)	4.8.3(c)
Coder, Tape MX-4495( )/GRA-71	3.13.2	4.9.2(j)
Coder, Tape MX-4496( )/GRA-71	3.14.2	4.10.2(c)
Keyer KY-468( )/GRA-71	3.15.2(c)®	4.11.2(c)
Keyer, Adapter MX-4498( )/GRA-71	3.16®	4.12

® Sample units of this item shall be checked for three (3) of the listed requirements. Samples units from the same lot shall be checked for different combinations of the specified requirements. All requirements shall be checked in any lot submitted.

4.18 Inspection for physical and functional interchangeability. (See 3.18).-

4.18.1 General.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.18.2 The following dimensional and operational checks shall be performed on Magazine, Recording Tape MA-9( )/GRA-71 using test gauges and fixtures fabricated in conformance with the configuration and dimensions of the Keyer KY-468/GRA-71.

(a) Check relation of tape and tape guides to cartridge attaching holes.

(b) Check locking plate movement. Plate must move freely with minimum clearance between the stop retainer tang and base plate. Maximum allowable clearance: 0.005 inch.

(c) Drive gear must extend 0.050 inch (± 0.001) over mounting face.

(d) Locking plate mounting studs shall not extend beyond flush position with mounting face.

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4.18.3 Check tape head track position azimuth and depth alinement on Keyer KY-468( )/GRA-71 using appropriate alignment gages and fixtures. A skewed head will cause compression of spaces between dots and dashes when a recorded message is played back into test set up, Figure 1.

4.19 Visual and mechanical inspection.- The units shall be examined for the defects in Standard MIL-STD-252 and the following:

4.19.1 Magazine, Recording Tape MA-9( )/GRA-71.- After assembly, but prior to attaching the case, inspect and test the MA-9/GRA-71 for the following items:

(a) All "E" rings, used as gear retainers shall be in place.

(b) All screws shall be tight and not burred.

(c) Gears shall be aligned and properly meshed with slight backlash.

(d) "E" rings shall not turn with gears.

4.19.2 Keyer, Adapter MX-4498/GRA-71.- After assembly, but prior to mounting in the case, inspect and test the MX-4498/GRA-71 for the following items:

(a) All screws and nuts shall be tight and not burred.

(b) Dessicant tube cover shall be in place and must fit tube tightly.

(c) Molded rubber pockets shall be positioned properly and well fitted.

(d) All cables shall be well mounted and connectors shall have appropriate caps attached properly.

(e) All connector pins shall be aligned.

(f) Water-tight gasket in case shall be well seated and free of dust, grease, etc. Trunk latches shall be solidly attached to case.

4.20 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Coder-Burst Transmission Group AN/GRA-71 shall be preserved and packaged in accordance with the requirements of Specification MIL-P-116, and as specified in the following paragraphs:

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-6, as follows; place each technical manual within a close-fitting water-proof bag, conforming to Specification MIL-B-117, heat seal closure.

5.1.1.2 Coder-Burst Transmission Group AN/GRA-71 shall be packaged Method IC-5, as follows; stow all components with the covering case, close the case and secure fastenings. Cushion the case on all surfaces with cells or pads of fiberboard, designed to absorb the shock of impact normally encountered in handling and transit. Place the cushioned case within a water-resistant fiberboard box, W5c or W6c, conforming to Specification PPP-B-636. Seal the entire closure, corners and joints, with water-resistant pressure-sensitive tape, conforming to PPP-T-76. Secure the technical manuals to the fiberboard box, with pressure-sensitive tape.

5.1.2 Level C.- Coder-Burst Transmission Group AN/GRA-71 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Coder-Burst Transmission Group AN/GRA-71 shall be packed within the following type shipping containers;

Container	Specification	Max Gross Wt.
Box, nailed wood	PPP-B-621	200 pounds
Box, wood cleated plywood	PPP-B-601	200 pounds
Box, fiberboard	PPP-B-636	Table II
Box, fiberboard, corrugated triple wall	PPP-B-640	200 pounds

Shipping containers shall be closed as specified in the appendix of the applicable specification. Wood shipping containers shall be strapped with metal strapping, fiberboard shipping containers shall be reinforced with bands of pressure-sensitive filament tape, conforming to PPP-T-97, metal strapping shall conform to QQ-S-781.

5.2.2 Level C.- Coder-Burst Transmission Group AN/GRA-71 shall be packed within shipping containers in a manner that will afford adequate protection against damage to the package and its contents during direct shipment to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.- Coder-Burst Transmission Group AN/GRA-71 is used for special purpose applications. It provides a means for preparation of coded messages and a facility for automatic keying of a transmitter.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Units required.

(c) Level of packaging and level of packing required for shipment. (Level A or level C.)

(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

6.3 Definitions of inspection terms.- The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference:

6.3.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.1.2 Testing.- Testing consists of determinations of compliance, using technical means.

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6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

6.5 Location of operational inspection.- It is desirable that the operational inspection (4.17) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspection sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRA-71W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army-EL

Preparing Activity
Army-EL

Project No. 5820-A282

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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